

Computational Astrophysics I

Assignment 2: χ^2 and KS test

Assignment Instructions

This assignment will be graded and must be submitted by **12th September, 2025**.

For both questions, you are expected to write your own functions; do not use the built-in `scipy` functions to perform the tests.

Please submit two files:

1. A Jupyter Notebook with your codes.
2. A single PDF containing your final answers, plots, and interpretations.

Email both these files to `saee.dhawalikar@iucaa.in` and `anirban.kopty@iucaa.in`, with a cc to `aseem@iucaa.in`. Use the subject line: **CA-I Assignment-I 2025**.

1 χ^2 Test

The file `chi2.csv` provided to you contains 3 data sets, A, B and C. Perform χ^2 test on each of these data sets, and interpret the results.

2 Kolmogorov–Smirnov (KS) test

- The file `ks.fits` contains 6 data sets, structured differently from the FITS files discussed in class. Examine its format and load the six arrays. If you are unable to read the file, you may use the function provided in `fits.py` to load the data.
- Perform the KS test on each of these 6 data-sets, comparing them with the standard normal distribution $\mathcal{N}(0, 1)$.
- Report your results interpretations, along with suitable visualizations.